

MAHNOOR REHMAN ABBASI

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Ajman, United Arab Emirates | IT Support Specialist, AVN Shipping FZE

PROFESSIONAL SUMMARY

Electronics Engineering graduate with a strong foundation in AI, machine learning, autonomous systems, and battery technologies, demonstrated through rigorous academic projects, certifications, and research experiences. Research interests include multimodal AI systems, sensor fusion in autonomous vehicles, and energy management in electric vehicles. Conducted undergraduate research in optical sensor systems and completed advanced certification projects involving simulations, algorithm implementation, and peer-reviewed reports. Seeking Master's studies to advance in AI-driven robotics and sustainable energy systems, with skills in predictive modeling, Kalman filters, and data-driven experimentation.

Core Competencies: Machine Learning · Deep Learning · NLP · Autonomous Systems · ESP32/STM32 Microcontrollers · Python · Predictive Modelling · Data Analytics · Chatbot Development · Cyber-Security

EDUCATION

Bachelor of Science in Electronics Engineering Sep 2020 – Sep 2024
Quaid-i-Azam University, Islamabad, Pakistan

- EQF Level 6 qualification

RESEARCH EXPERIENCE AND PROJECTS

AI and Machine Learning Research (Microsoft AI & ML Engineering Specialization and Certifications) 2024–2025

- **Capstone Project:** Developed a predictive model (e.g., fraud detection system or intelligent chatbot) involving data preparation, deep learning training, deployment on Azure, and evaluation for scalability and bias mitigation.

Autonomous Vehicles and Robotics Research (University of Toronto Self-Driving Cars Specialization and University of Michigan Certification) 2024–2025

- **Projects in CARLA Simulator:** Programmed vehicle control models; implemented extended Kalman filters for sensor fusion (GPS, IMU, LIDAR); performed feature matching and CNN-based object detection/semantic segmentation; integrated perception with planning for a full autonomous software stack.

Battery Technologies and Management Systems Research (Arizona State University Battery Technologies Specialization) 2024–2025

- **Supporting Analysis:** Evaluated trade-offs in battery types (Li-ion, Zn/Ni) for EV applications; foundational circuit design in Georgia Tech's Introduction to Electronics, including op-amp filters and diode regulators.

LifeLens – Multimodal Emotion Detection System

- LifeLens is an advanced AI-driven system designed for real-time emotion recognition by fusing multiple data modalities, including facial expressions, voice tone, and textual input. Leveraging deep learning architectures such as convolutional neural networks (CNNs) for visual features and recurrent models for audio-text sequences, the project achieved robust classification across emotions like happiness, anger, sadness, and surprise.

SmartSplit – Intelligent Expense Management Application

- SmartSplit is an AI-powered mobile/web application that automates personal and group expense tracking, categorization, and bill splitting. Utilizing machine learning models (e.g., supervised classification for transaction labeling and predictive analytics for spending forecasts), it processes receipts via OCR, detects patterns in user data, and suggests fair splits among groups.

IVAMS – Intelligent Vehicle Automation and Monitoring System

- IVAMS is an ESP32 microcontroller-based prototype for autonomous vehicle monitoring and automation, incorporating real-time sensor data acquisition (e.g., temperature, GPS, accelerometers), control mechanisms, and Vehicle-to-Everything (V2X) communication concepts.

PROFESSIONAL CAREER

IT Support Specialist **Oct 2025 – Present**
AVN Shipping FZE, Ajman, UAE

- Provide comprehensive technical support and troubleshoot hardware/software issues for logistics operations

Data Analyst (Remote) **Aug 2025 – Present**
Lotus BPM AI, Texas, USA

- Perform data analysis and develop AI-driven solutions for business process optimization

Proxy Generating Agent (AI Content Strategist) **Sep 2024 – Sep 2025**
Hackta Connect (Remote)

- Designed AI-driven content strategies and produced 300+ high-quality blogs using generative AI

Artificial Intelligence Engineer (Remote) **Aug 2021 – Jan 2025**
Overpass

- Implemented machine learning algorithms for predictive analytics and data-driven decision making

Chatbot Developer (Freelance) **Aug 2023 – Jan 2024**
Upwork

- Built NLP-powered conversational chatbots for healthcare, e-commerce, and education clients

Prompt Engineer (Internship) **Aug 2023 – Jan 2024**
Elite International, Islamabad, Pakistan

- Engineered prompts for NLP systems and automation workflows

Undergraduate Researcher **Nov 2021 – Dec 2021**
Quaid-i-Azam University

- Conducted research on control mechanisms, and Vehicle-to-Everything (V2X) communication concepts.

TECHNICAL & EXPERTISE

Programming & Tools: Python (Advanced), C++, MATLAB, Microsoft Excel (Advanced), Microsoft Office Suite

AI/ML Technologies: Machine Learning, Deep Learning, Neural Networks, Natural Language Processing, Prompt Engineering, Predictive Modelling, Data Analytics, TensorFlow, PyTorch

Autonomous Systems: ESP32/STM32 Microcontrollers, Real-time Sensing & Control, V2X Concepts, Self-Driving Car Technologies, Sensor Integration

Electric Vehicles: Battery Technologies (Li-based, Zn/Ni-based), EV Systems, Energy Management

Other Technical: Data Security, Dashboard Development, Chatbot Development, Technical Writing, Web Development

Creative & Media: Video Editing (Premiere Pro, CapCut), Graphic Design (Canva, Photoshop, Illustrator), Digital Content Creation

CONFERENCES & WORKSHOPS

- **International Workshop on Data and AI-Driven Mineral Exploration Quaid-i-Azam University (QAU), Islamabad (Aug 2025)**– Focused on AI applications in geoscientific data analysis and mineral exploration
- **Three-Day Training Workshop on AI & Data Sciences - Advancing Research Across All Sciences**– Pakistan Scientific and Technological Information Center (PASTIC) (2023–2025)
- **Seminar on Emerging Role of Artificial Intelligence**– Institute of Information Technology (IIT), QAU in collaboration with ORIC (Dec 2024)
- **Seminar on "Humans & Machines"**– IIT Society, QAU (Dec 2024) – Explored symbiotic relationship in innovation
- **Workshop on AI and Technology in 2050**– Institute of Information Technology, QAU (Nov 2023)
- **Training on SPSS, AI, and Data Science**– Quaid-i-Azam University (2023–2024) – Hands-on departmental training sessions
- **Innovation Summit for Microelectronics and Artificial Intelligence**– Department of Electronics, QAU in collaboration with IRP (Jun 16–17, 2021) – Online event featuring international speakers from Pakistan, China, and Italy

ADMINISTRATIVE ACTIVITIES

- **Fundraising Manager** – Feeding People Foundation (Children) – Successfully raised funds exceeding \$5,000 for annual charity initiatives supporting children in need
- **AI Ambassador / Volunteer Tutor** (Jul–Aug 2025) – Mentored students in AI concepts and practical applications
- **Digital Media & Social Media Management** – Enhanced brand visibility through strategic content creation
- **Research Administration Support** – Assisted in lab documentation and photonics sensor experiments

CERTIFICATIONS

- Neural Networks and Deep Learning – DeepLearning.AI (Mar 2025)
- AI and Machine Learning Algorithms and Techniques – Microsoft (Jun 2025)
- Microsoft AI & ML Engineering – Microsoft (Mar 2024)
- Microsoft Azure Machine Learning – Microsoft (Aug 2025)

- Google AI Essentials – Google (Apr 2025)
- Big Data Essentials – Google (Feb 2025)
- Self-Driving Cars Specialization (4 courses) – University of Toronto (Jan–Feb 2025)
- AI for Autonomous Vehicles and Robotics – University of Michigan (Feb 2025)
- Batteries and Electric Vehicles (Series) – Arizona State University (Jul 2025)
- Microsoft IT Support Specialist – Microsoft (Jul 2025)
- Google IT Support & Google Cybersecurity – Google (2024–2025)
- Python – Great Learning (Jul 2025)
- Introduction to Electronics – Georgia Institute of Technology (Apr 2025)
- Google Prompting Essentials – Google (Jul 2025)
- Work Smarter with Microsoft Excel – Microsoft (Jul 2025)
- SEO Fundamentals – SEMrush Academy (Feb 2024)

LANGUAGES

English (Fluent) · Urdu (Native) · Arabic (Intermediate) · Punjabi (Native)

REFERENCES

Dr. Qurrat-ul-Ain Minhas – Professor
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Dr. Muhammad Zia – Professor
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